Monday 11:30 – 12:30 ⇒ ESB-323

Engineering Circuit Analysis : Hayt & Kemmeny

Current: Rate of flow of charge at a given cross-section.

Power: $v.i$ ($v \rightarrow$ potential)

$P>0 \longrightarrow$ dissipated

$P<0 \longrightarrow$ generated

Lumped Element: Only v and i is enough to find Resistance

Circuit: a path in a graph that start and ends at the same vertex, with no repeated edges and atleast one edge. (Actual graph theory)

Kirchoff's Laws:

Voltage law: Take any loop with elements ①, ②, ③, ..., ⓝ, ① ; $\sum_{i=1}^{n} v_i = 0$

(Analogos to $\oint \bar{E}.d\bar{L} = 0$ (if no magnetic flux)

Current law: Take any node 'k' and look at the elements 1, 2, ..., m connected to it.

Total current entering k - total current leaving k

total current entering = total current leaving

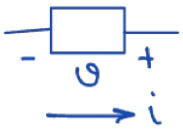
(No accumulation of charge ; perfect conductor)

(Analogos to Gauss law)

These 2 laws have nothing to do with resistance or whatever, it can be applied to any circuit

Linear Element: V-I has linear relationship

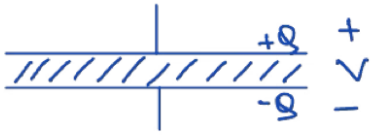
(Resistane, capacitors, inductors etc....)



Element with linear i-v characteristic is a straight line through origin (Recall the definition of linearity from Signals and Systems

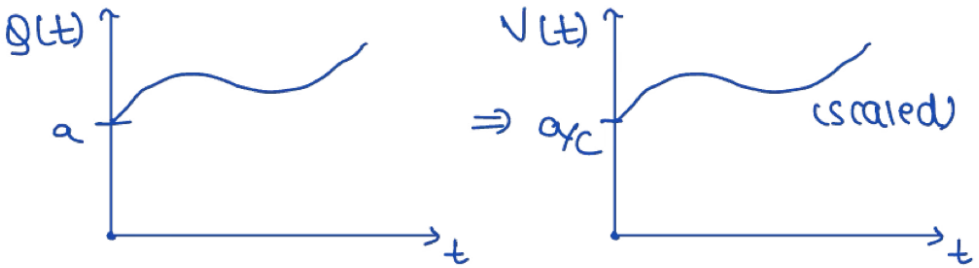
$x_1 \to y_1$; $x_2 \to y_2 \Rightarrow ax_1+bx_2 \to ay_1+by_2$; $a,b \in \mathbb{R}$)

Capacitor:



$Q = CV$

C → capacitance (Farad)



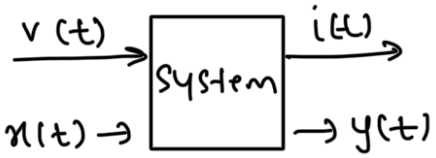
$$I(t) = \frac{d}{dt} Q(t) \quad \Rightarrow \quad I(t) = C \frac{dV(t)}{dt}$$

$V_1(t) \to I_1(t)$; $V_2(t) \to I_2(t)$

$$\alpha_1 V_1(t) + \alpha_2 V_2(t) \longrightarrow C\frac{d}{dt}(\alpha_1 V_1(t) + \alpha_2 V_2(t))$$

$$\longrightarrow \alpha_1 C\frac{dV_1}{dt} + \alpha_2 C\frac{dV_2}{dt}$$

$$\longrightarrow \alpha_1 I_1(t) + \alpha_2 I_2(t)$$



linear system: $\alpha_1 x_1 + \alpha_2 x_2 \downarrow \alpha_1 y_1 + \alpha_2 y_2$

Capacitor system follows it (linearity)

If input is $i(t)$ and output is $V(t)$

$$I(t) = C\frac{dV(t)}{dt} \Rightarrow V(t) = \frac{1}{C}\int_0^t I(u)\,du + V(0)$$

$I_1(t) \to V_1(t)$; $I_2(t) \to V_2(t)$

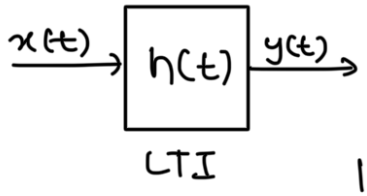
$$\Rightarrow \alpha_1 I_1(t) + \alpha_2 I_2(t) \longrightarrow \frac{1}{C}\int_0^t (\alpha_1 I_1(u) + \alpha_2 I_2(u))\,du + V(0)$$

$$= \alpha_1 V_1(t) + \alpha_2 V_2(t) + \underbrace{V_0 - \alpha_1 V_1(0) - \alpha_2 V_2(0)}$$

This linear iff the initial voltage is zero

Capacitor is not exactly linear, but almost linear

{ resistor + battery/source }



$$y(t) = \int x(u)\,h(t-u)\,du$$

In Laplace-domain:

(1-sided Laplace) $\Rightarrow X(s) = \int_0^\infty e^{-st} x(t)\,dt$ $\quad s \rightarrow$ complex

$$X(s) \xleftrightarrow{L} x(t); \quad H(s) \xleftrightarrow{L} h(t); \quad Y(s) \xleftrightarrow{L} y(t)$$

$$\Rightarrow Y(s) = H(s)\,X(s) \quad \text{for LTI}$$

Take Resistor for now:

$$V(t) = Ri(t) \Rightarrow V(s) = R\,I(s)$$

$H(s)$ is a constant function (called gain)

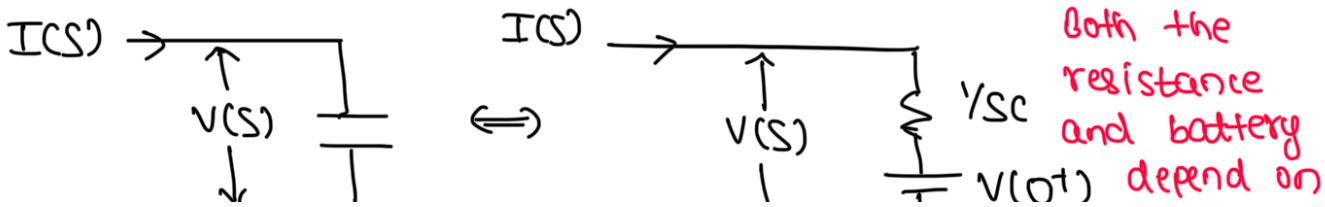
Now we see capacitor:

$$I(t) = C\frac{dv(t)}{dt} \Rightarrow I(s) = C\,\mathcal{L}\left(\frac{dv(t)}{dt}\right)$$

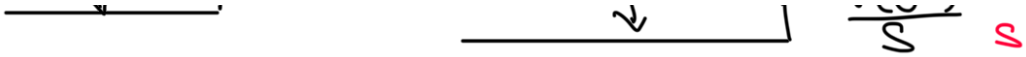
$$= C[sV(s) - v(0^+)]$$

$$= CsV(s) - Cv(0^+)$$

$$\Rightarrow V(s) = 1/sC\; I(s) + \frac{v(0^+)}{s}$$



Both the resistance and battery depend on



(Almost linear)

Circuit with resistor / cap / battery

⇕

circuit with resistor / battery in Laplace Domain

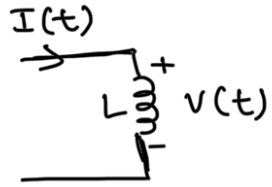
⇒ Solve KVL, KCL in s-domain, then laplace inverse.



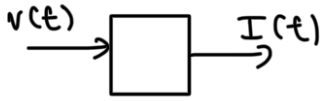
ideal voltage source

Inductor:

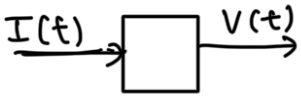
$$v(t) = L\frac{dI(t)}{dt}$$



$$\phi = -L.I(t) \; ; \; V = -\frac{d\phi}{dt} = L\frac{dI(t)}{dt}$$



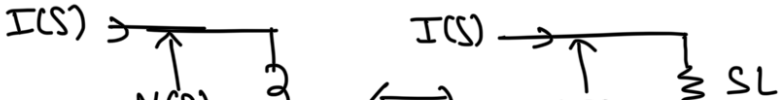
this will be linear iff $I(0) = 0$

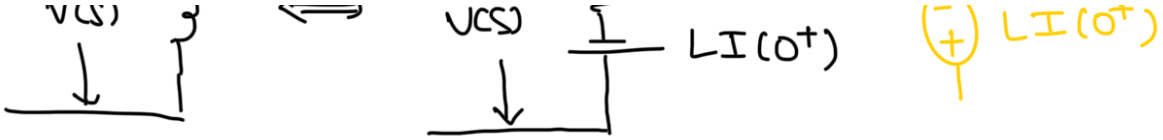


this will be linear here always

we go to laplace domain directly:

$$V(S) = SLI(S) - LI(0^+)$$





Capacitor & Inductor: "almost linear"

↳ in t-domain

Capacitor: $\frac{1}{SC}$(Res) + $\frac{v(0)}{S}$

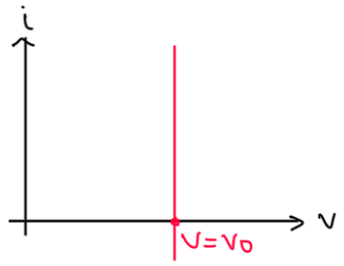
Inductor: SL(Res) + LI(0)

↳ in s-domain = Resistor is s-dependent + Voltage source

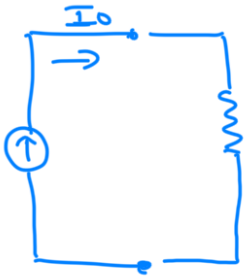
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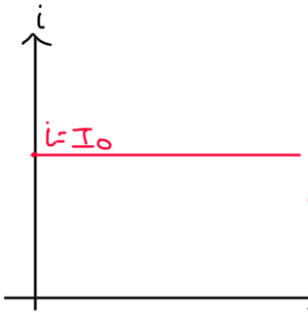
ideal voltage source: I'll fix a voltage and draw any amount of current (upto infinite)

ideal current source: provides a fixed current (I_0) irrespective of the voltage across it.



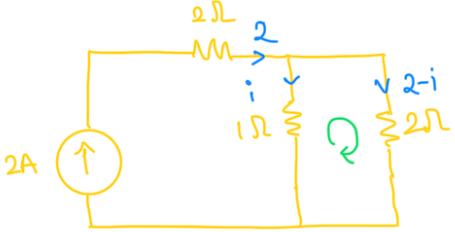
zero resistance





infinite resistance

Do not write KVL with current source in a loop



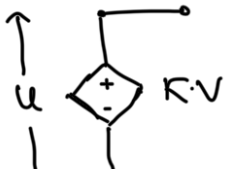
KVL: $(2-i)2 - i = 0 \Rightarrow i = 4/3$

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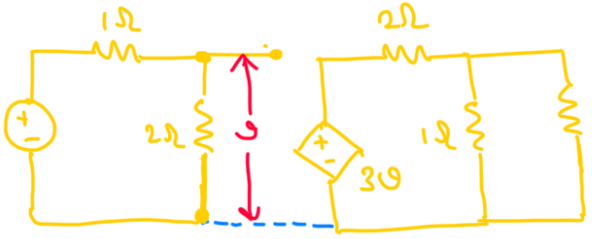
controlled source:~

aka dependent source

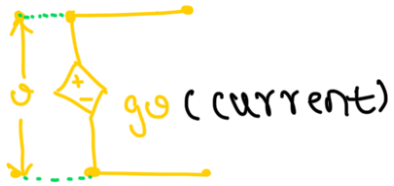
2 terminal (control)



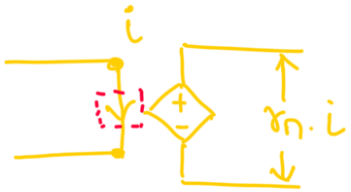
+ 2terminal (source)

① 

→ voltage controlled voltage source (VCVS)

② VCCS ⇒ 



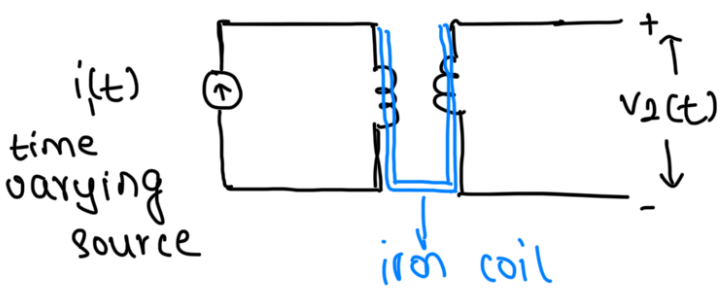
③ CCVS ⇒ 

④ CCCS ⇒ 

if i or k or g is -ve, we flip the controlled source appropriately.

Both CCVS & VCCS have unit-change
Follow one-convention for sign-convention.

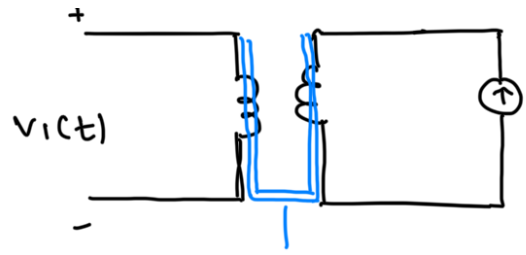
Mutual Inductance:



$v_2(t) \propto \frac{di_1(t)}{dt}$

(Since $\phi_2(t) \propto i_1(t)$)

constant of proportionality = M_{21}

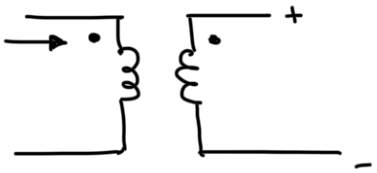


$v_1(t) \propto \frac{di_2(t)}{dt}$

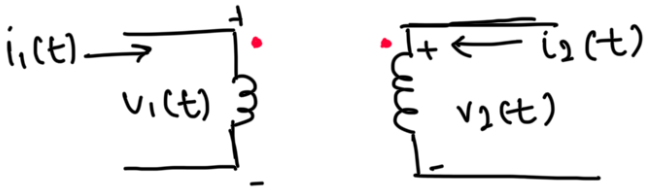
constant of proportionality $= M_{12}$

$M_{21} = M_{12} = M$ (Symmetry principle / Energy conservation)

How to decide sign? (dot convention)



"A current (increasing) entering dot side of a coil creates a +ve voltage on the dot side of the other coil."

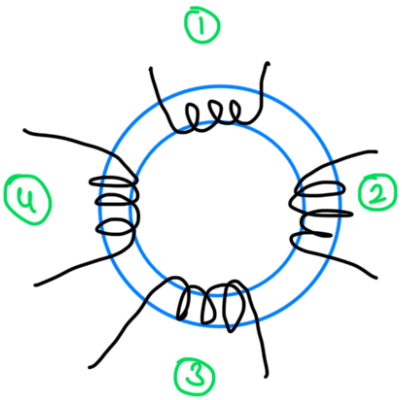


$v_1(t) = L_1 \frac{di_1(t)}{dt}$ if $i_2(t) = 0$

$v_2(t) = L_2 \frac{di_2(t)}{dt}$ if $i_1(t) = 0$

$$v_1(t) = L_1 \frac{di_1(t)}{dt} + M \frac{di_2(t)}{dt} \qquad v_2(t) = L_2 \frac{di_2(t)}{dt} + M \frac{di_1(t)}{dt}$$

$$\begin{bmatrix} L_1 & M_{12} & M_{13} \cdots & M_{1n} \\ M_{21} & L_2 & M_{23} \cdots & M_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ M_{n1} & M_{n2} & M_{n3} & L_n \end{bmatrix} = M$$



$$\Rightarrow \begin{bmatrix} v_1(t) \\ v_2(t) \\ \vdots \end{bmatrix} = M \begin{bmatrix} di_1(t)/dt \\ di_2(t)/dt \\ \vdots \end{bmatrix}$$

($M \rightarrow$ symmetric matrix)

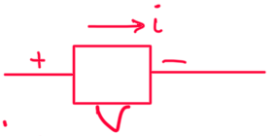
$\begin{bmatrix} \vdots \\ v_n(t) \end{bmatrix}$ $\begin{bmatrix} \vdots \\ di_n(t)/dt \end{bmatrix}$

The equation above in red, convert into laplace:

$V_1(s) = L_1(sI_1(s) - I_1(0)) + M_{12}(sI_2(s) - I_2(0))$

$V_2(s) = \ldots$

$$\Rightarrow \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = s\begin{bmatrix} L_1 & M_{12} \\ M_{12} & L_2 \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} - \begin{bmatrix} L_1 & M_{12} \\ L_2 & M_{21} \end{bmatrix}\begin{bmatrix} I_1(0) \\ I_2(0) \end{bmatrix}$$



$V \cdot i > 0 \Rightarrow$ absorbing power/energy

$V \cdot i < 0 \Rightarrow$ Releasing power/energy for resistance;

voltage drop is always along current $\Rightarrow$ always absorbing

Capactior: charging $\rightarrow$ absorbs energy ($V \le 0, \downarrow$) ($V \ge 0, \uparrow$)

discharging $\rightarrow$ releasing energy ($V \ge 0, \downarrow$) ($V \le 0, \uparrow$)

Inductor: $i \ge 0, \uparrow$; $i \le 0, \downarrow \Rightarrow$ absorbtion

$i \le 0, \uparrow$; $i \ge 0, \downarrow \Rightarrow$ releasing

$$\text{Energy}(t) = \int_0^t P(\tau)\,d\tau \begin{cases} \ge 0 & \text{(absorbing)} \\ \le 0 & \text{(releasing)} \end{cases}$$

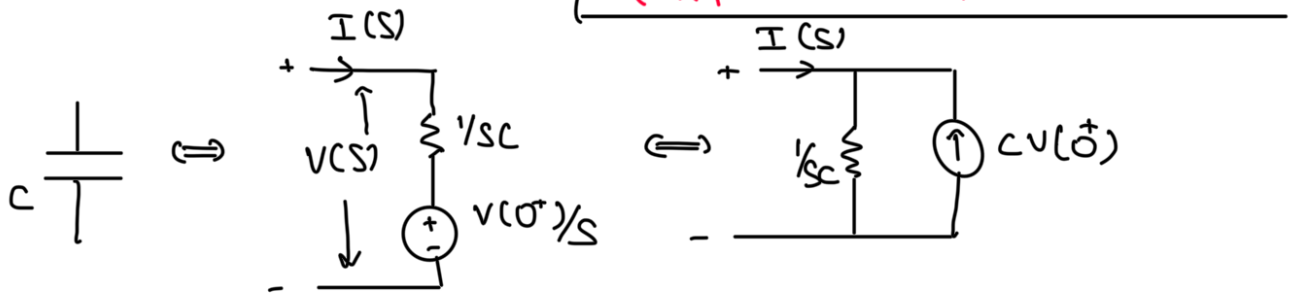
Capacitor: $E(t) = \frac{1}{2}C\left[(V(t))^2 - (V(0))^2\right]$

(energy depends on final

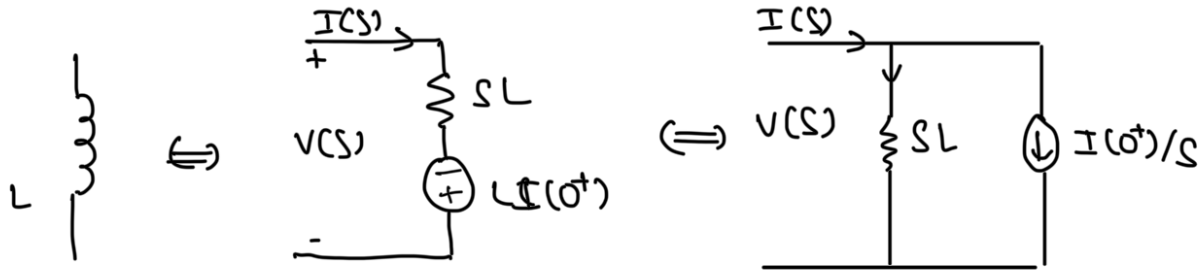
(energy depends on final state)

Inductor: $E(t) = \frac{1}{2} L [(I(t))^2 - (I(0))^2]$

(depends on final current)

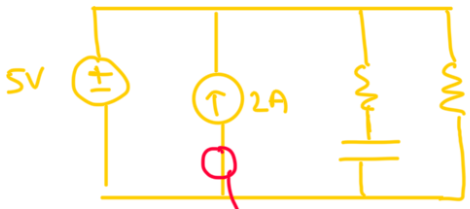


$$I(s) = sC\left\{V(s) - \frac{V(0^+)}{s}\right\} \Rightarrow I(s) = sC\,V(s) - C\,V(0^+)$$



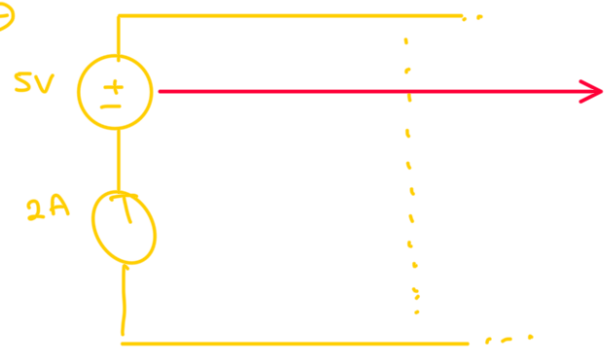
→ two voltage sources in parallel
→ two current sources in series
} not allowed

→



having this current source or not doesn't matter since voltage source can generate infinite current so the 2A doesn't matter

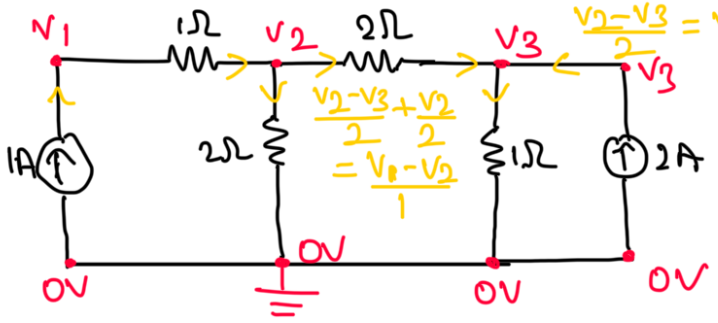
→



this doesn't matter as current source can have ∞ voltage.

Nodal Analysis: use KCL with help of node voltage

Nodal Analysis: use KCL with help of node voltage



$I = \frac{V_1 - V_2}{1}$

$\frac{V_2 - V_3}{2} + \frac{V_2}{2} = \frac{V_1 - V_2}{1}$

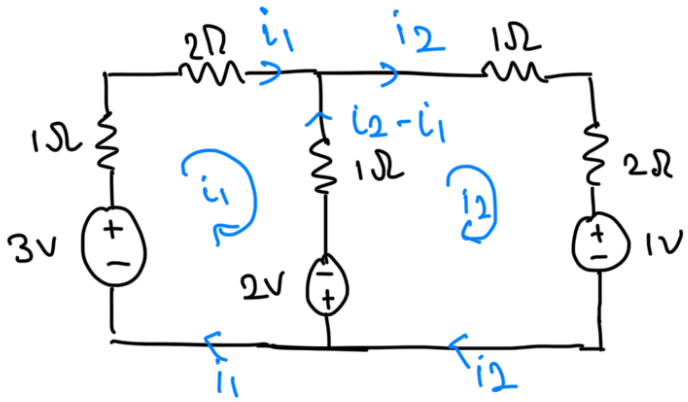
$\frac{V_2 - V_3}{2} = \frac{V_3}{1} + 2$

$V_3 = -6/5$

$V_2 = 2/5$

$V_1 = 7/5$

Mesh Analysis: use KVL + "mesh-current"



mesh

Go for small loops that doesn't contain any other loops

$-3 + i_1 + 2i_1 - (i_2 - i_1) - 2 = 0 \Rightarrow 4i_1 - i_2 = 5 \Rightarrow i_2 = -7/15$

$+2 + (i_2 - i_1) + i_2 + 2i_2 + 1 = 0 \Rightarrow -i_1 + 4i_2 = -3 \quad i_1 = 41/30$
